

Worked Example 2: Economic Production Quantity

$A = 450, D = 12,000/\text{yr}, v = 18.5, r = 0.22, m = 48,000/\text{yr}$

Inventory and Production Management

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Model

Assumption 8 of the EOQ is relaxed: the lot is produced at a finite rate $m > D$.

$$TRC(Q) = \frac{AD}{Q} + \frac{Q(1 - D/m)vr}{2}$$

$$EPQ = \sqrt{\frac{2AD}{vr(1 - D/m)}} = \frac{EOQ}{\sqrt{1 - D/m}} = 1,880.98$$

$$I_{max} = Q(1 - D/m) = 1,410.7$$

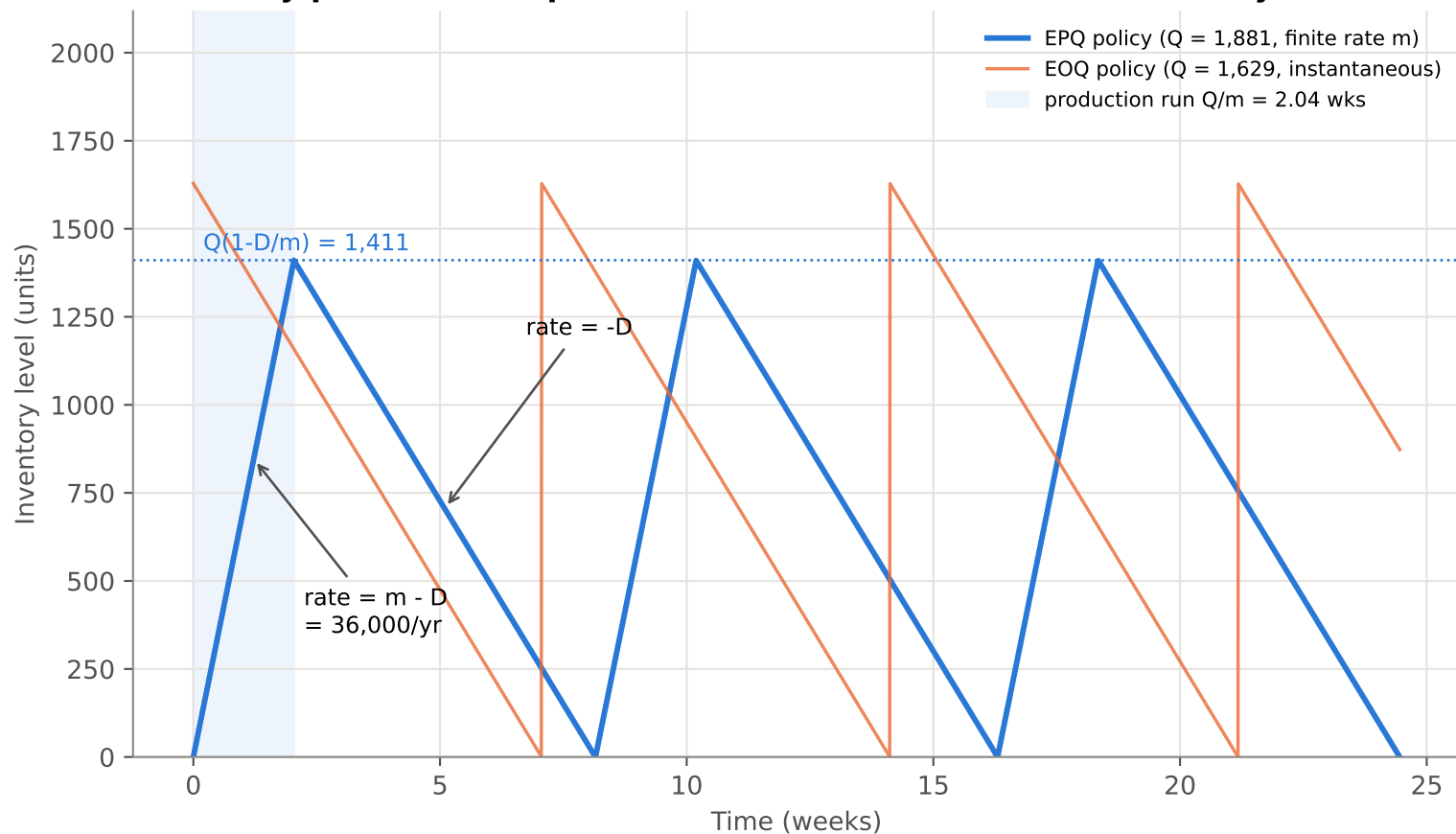
Interpretation

Produce 1,881 units per run, 6.38 times per year. Each run lasts 2.04 weeks and the line is then free for 6.11 weeks for other products.

Numerical results

Quantity	Value
EPQ (units)	1,881
EOQ for comparison (units)	1,629
$EPQ / EOQ = 1/\sqrt{1-D/m}$	1.155
TRC(EPQ) (\$/yr)	5,742
Setup cost AD/Q (\$/yr)	2,871
Carrying cost (\$/yr)	2,871
Maximum inventory $Q(1-D/m)$ (units)	1,411
Average inventory (units)	705.4
Production runs per year D/Q	6.38
Cycle length Q/D (weeks)	8.151
Production run length Q/m (weeks)	2.038
Line idle time per cycle (weeks)	6.113
Utilisation of the line D/m	0.25

Inventory profile: finite production rate vs instantaneous delivery



EPQ/EOQ = $1/\sqrt{1-D/m} = 1.155$

